Exercise Sheet 17

Discrete Mathematics by Hongfei Fu, 2023.12.6

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1. Show that a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.

   Answer Area:

   - Suppose that a simple graph G $= V, E$ is a tree.
     $\because$ The graph is a tree.
     $\therefore$ The graph is connected.
     For an arbitrary edge $e \in E$, assuming its endpoints are v_1 and v_2. Since there is only one unique path between v_1 and v_2, which is e, v_1 and v_2 are not connected after the edge e is removed. That is to say, the graph is not connected.
     $\therefore$ G is connected but the deletion of any of its edges produces a graph that is not connected.
   - Suppose that a graph G $= V, E$ is connected and the deletion of any of its edges produces a graph that is not connected.
     Assuming that there is a simple circuit in the graph and the edge e is in the circuit.
     Then after removing the edge e, the graph is still connected since there is a new path between any two vertices by replacing e in the former path (if there is) with the remaining part of the circuit.
     Therefore, here is a contradiction. The assumption is wrong.
     Therefore there is not a simple circuit in the graph.
     $\because$ G is connected.
     $\therefore$ G is a tree.

   In conclusion, a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.

2. (**2pt**) Let G be a simple graph with n vertices. Show that G is a tree if and only if G has no simple circuits and has $n-1$ edges. [Hint: To show that G is connected if it has no simple circuits and $n-1$ edges, show that G cannot have more than one connected component.]

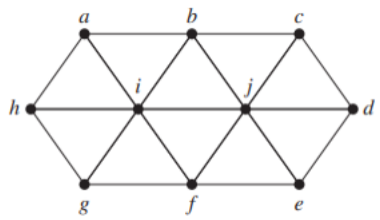
   Answer Area:

   - If $G = V, E$ is a tree.
     Then G has no simple circuit and $|E| = |V| - 1 = n - 1$.
   - If G has no simple circuits and has $n-1$ edges. Assuming that G is not connected, that is, G has s connected components: $G_1 = (V_1, E_1), G_2 = (V_2, E_2), \ldots, G_n = (V_s, E_s)$.
     Every connected component is a tree since it has no simple circuit.
     Therefore, $|E| = n - 1 = \sum_{k=1}^{s} |E_k| \leq \sum_{k=1}^{s}(|V_k| - 1) = |V| - s = n - s$.
     Therefore $s \leq 1$. So s = 1.
     Therefore G is connected. Hence, G is a tree.

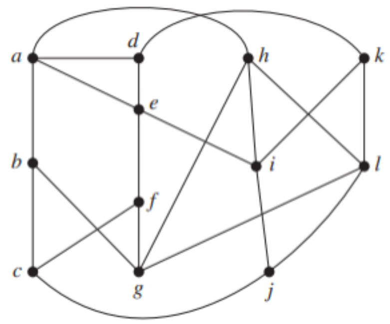
   In conclusion, G is a tree if and only if G has no simple circuits and has $n-1$ edges.

3. Find a spanning tree of the following graphs.

a)

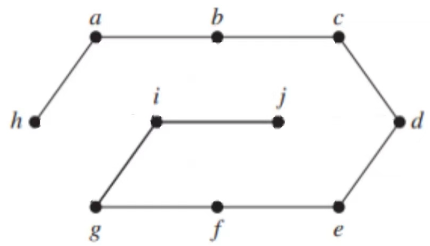


b)



Answer Area:

a)



b)